

**Longitudinal Graph-Transformer Network for
Personalised Prediction of Mild Cognitive
Impairment to Alzheimer's Disease Conversion**

Dissertation

COMP702 — M.Sc. Project (2025/26)

Submitted by

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Student Declaration

I confirm that I have read and understood the University's Academic Integrity Policy.

I confirm that I have acted honestly, ethically and professionally in conduct leading to assessment for the programme of study.

I confirm that I have not copied material from another source nor committed plagiarism nor fabricated data when completing the attached piece of work. I confirm that I have not previously presented the work or part thereof for assessment for another University of Liverpool module. I confirm that I have not copied material from another source, nor colluded with any other student in the preparation and production of this work.

I confirm that I have not incorporated into this assignment material that has been submitted by me or any other person in support of a successful application for a degree of this or any other university or degree-awarding body.

SIGNATURE: Tamizhan Elango

DATE: 11th September 2026

Dedication

This dissertation is dedicated to my family, whose unwavering support and encouragement throughout my MSc journey made this work possible.

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Dr Blaine Keetch, for his guidance, constructive feedback, and support throughout the duration of this project. His critical questions during the proposal review sharpened the technical rigour of this work considerably.

I am grateful to the Alzheimer's Disease Neuroimaging Initiative (ADNI) for providing access to the longitudinal clinical dataset that made this research possible. Data collection and sharing for ADNI is funded by the National Institute on Aging and the National Institute of Biomedical Imaging and Bioengineering.

I would also like to thank the School of Computer Science and Informatics at the University of Liverpool for providing the computational resources and academic environment that supported this research.

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Abstract

Alzheimer's Disease (AD) is the most prevalent cause of dementia worldwide, affecting more than 55 million people globally. Mild Cognitive Impairment (MCI) represents the earliest clinically detectable precursor to AD, yet existing artificial intelligence systems largely address the problem as a static, single-visit classification task — predicting whether a patient has MCI or AD at a given moment, rather than predicting when a patient with MCI will progress to AD. This dissertation addresses that gap.

This work presents LGTP, the Longitudinal Graph-Transformer for Personalised Prognostics, a novel deep learning framework that predicts the time-to-conversion from MCI to AD using longitudinal clinical data from the Alzheimer's Disease Neuroimaging Initiative (ADNI). LGTP integrates three components not previously combined in a single system: a Transformer encoder that models each patient's sequence of clinical visits over time; a Graph Attention Network (GATv2) that enriches patient representations using population-level information from clinically similar patients; and a DeepSurv survival analysis output head that produces personalised conversion probability curves at 12, 24, and 36 months rather than a binary diagnostic label. The framework was trained and evaluated on a cohort of 1,053 MCI patients (401 converters, 652 stable) drawn from 8 merged ADNI CSV files.

Results demonstrate that LGTP achieves a concordance index (C-index) of 0.865 and a time-dependent AUC of 0.905 at 24 months and 0.930 at 36 months on the held-out test set of 160 subjects, with an Integrated Brier Score of 0.154 confirming well-calibrated predictions. The Expected Calibration Error of 0.018 demonstrates that the model's confidence estimates are reliable, with correctly classified patients exhibiting $3.6\times$ lower epistemic uncertainty than misclassified ones. CDR-SB was identified as the dominant predictive biomarker via SHAP attribution, consistent with its established role as the primary endpoint in Alzheimer's clinical trials. An ablation study confirmed the independent contribution of both the longitudinal sequence component (+0.054 C-index over baseline-only) and the population graph component. These results are competitive with state-of-the-art ADNI literature despite using only clinical cognitive assessment scores — without PET imaging, CSF biomarkers, or raw MRI — representing a meaningful step towards clinically trustworthy, personalised, and interpretable prognosis for neurodegenerative disease.

Statement of Ethical Compliance: A0

Data Category: A Participant Category: 0

I confirm that I have read the University's ethical guidelines and followed them throughout this project. This project uses only secondary, de-identified, open-access research data from the ADNI database under an approved Data Use Agreement. No human participants, surveys, interviews, or user trials were conducted at any stage of the project. All data is handled in compliance with the ADNI Data Use Agreement, which prohibits redistribution and re-identification of participants.

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Chapter 1: Introduction and Background

1.1 Background and Motivation

Neurodegenerative diseases represent one of the most significant and growing challenges in global healthcare. Alzheimer's Disease (AD), the most common form of dementia, affects an estimated 55 million people worldwide, a number projected to rise to 139 million by 2050 as populations age [1]. In the United Kingdom alone, dementia affects approximately 900,000 people and costs the economy an estimated £34.7 billion annually [2].

Mild Cognitive Impairment (MCI) is widely recognised as the earliest clinically detectable stage preceding AD. Individuals with MCI experience measurable cognitive decline — in memory, attention, or executive function — without meeting the threshold for dementia diagnosis. Crucially, MCI is a heterogeneous condition: approximately 38% of MCI patients progress to AD within three years (progressive MCI, or pMCI), while the remainder remain stable or even revert to normal cognition (stable MCI, or sMCI) [3]. This heterogeneity makes MCI-to-AD conversion prediction both clinically critical and scientifically challenging.

Early and accurate identification of which MCI patients are likely to convert — and within what timeframe — has profound clinical implications. It determines who should be prioritised for disease-modifying therapy, enrolled in clinical trials, monitored more intensively, or counselled about future care planning. Current clinical practice relies on periodic cognitive assessments, neuroimaging, and biomarker analysis, but these cannot reliably distinguish pMCI from sMCI at baseline, and no AI system currently provides clinicians with a personalised, time-aware risk estimate alongside a measure of confidence.

1.2 Limitations of Existing Approaches

Most existing deep learning systems for Alzheimer's diagnosis address a fundamentally different — and simpler — problem: cross-sectional classification. Given a patient's data at a single visit, these systems predict whether the patient is CN (cognitively normal), MCI, or AD. Prominent examples include CNN-based MRI classifiers, Vision Transformer architectures, and multimodal fusion systems such as the Uncertainty-Aware Multimodal Transformer (UAMT) developed as prior undergraduate work by this author [4].

These cross-sectional systems have three fundamental limitations for the conversion prediction problem. First, they treat each visit in isolation, discarding the temporal trajectory of cognitive decline which is itself a powerful predictor [5]. A patient whose MMSE score has fallen from 28 to 22 over 24 months carries very different risk than a patient who has maintained 22 over the same period. Second, they treat patients as independent entities, ignoring the information available from population-level patterns — the fact that a patient resembles 200 prior converters is itself a strong signal. Third, they produce a class label — AD or not AD — rather than a time-to-conversion estimate, which is what clinicians actually need for planning purposes.

A 2025 systematic review of deep learning models for MCI-to-AD conversion covering 68 studies identified multimodal fusion depth and longitudinal progression modelling as critical open gaps in the field [6]. A 2026 JMIR systematic review confirmed that while multimodal AI for AD diagnosis has advanced considerably, longitudinal and survival-based approaches remain underexplored [7].

1.3 Problem Statement

This project addresses the following research problem:

Given a patient's longitudinal sequence of clinical visits — including cognitive assessment scores, demographic information, and genetic markers — predict the probability that the patient will convert from MCI to Alzheimer's Disease at 12, 24, and 36 months from baseline, with calibrated confidence and a clinician-interpretable explanation of which features most influenced the prediction.

The key distinction from prior work is the shift from static diagnosis (what is this patient's status?) to dynamic prognosis (when will this patient convert?), and from opaque predictions to uncertainty-aware, explainable outputs.

1.4 Aims and Objectives

The principal aim of this project is to develop LGTP, a Longitudinal Graph-Transformer for Personalised Prognostics, that predicts MCI-to-AD conversion using the ADNI longitudinal dataset.

The specific objectives are:

1. To load, merge, and preprocess 8 ADNI CSV files into a unified longitudinal patient dataset with survival analysis labels.
2. To develop a Transformer-based temporal encoder that models each patient's visit sequence, capturing cognitive trajectory over time.
3. To build a population graph using patient-level similarity and process it with a Graph Attention Network (GATv2) to enrich individual patient embeddings.
4. To implement a DeepSurv survival analysis output head producing personalised time-to-conversion probability curves.
5. To quantify prediction uncertainty using Monte Carlo Dropout and generate SHAP-based biomarker attribution for each prediction.
6. To evaluate the framework using C-index, time-dependent AUC, Brier score, and a rigorous ablation study.

1.5 Overview of the Dissertation

Chapter 2 reviews the related literature across four areas: multimodal AD diagnosis, longitudinal modelling, graph-based patient modelling, and uncertainty quantification. Chapter 3 presents the design of the LGTP framework. Chapter 4 describes the implementation in detail. Chapter 5 presents experimental results and evaluation. Chapter 6 covers project ethics. Chapter 7 presents the conclusion and future work. Chapter 8 addresses BCS criteria and self-reflection.

Chapter 2: Related Work

2.1 Cross-Sectional Multimodal Diagnosis

The dominant paradigm in AI-based Alzheimer's research has been cross-sectional classification: training models on data from a single visit to distinguish CN, MCI, and AD. Early work used CNNs applied directly to structural MRI, achieving AUC values of 0.80-0.93 for AD versus CN classification [8]. The introduction of attention mechanisms and Transformer architectures improved performance further; a 2024 multimodal Transformer study by Yu et al. achieved state-of-the-art results by fusing MRI, PET, and clinical scores using cross-modal attention [9].

A 2025 multi-view multimodal deep learning framework (ADMV-Net) demonstrated that combining structural MRI with cognitive and demographic data via multi-view fusion and graph convolutional layers outperformed single-modality baselines across both AD-versus-CN and MCI-versus-CN classification tasks [10]. These cross-sectional approaches, while mature, do not address the conversion prediction problem.

2.2 Longitudinal Progression Modelling

Incorporating temporal information substantially improves conversion prediction. Kim et al. demonstrated that an LSTM model using longitudinal multimodal ADNI data achieved AUC 0.93 for 24-month MCI-to-AD prediction, significantly outperforming cross-sectional random forest baselines (AUC 0.87) [11]. A 2025 study introduced a Transformer architecture trained directly on longitudinal visit sequences from ADNI, demonstrating superiority over LSTM and GRU baselines as the number of prior visits increased [12].

Survival analysis provides a natural framework for conversion prediction as it handles right-censored observations — patients who did not convert during follow-up — without excluding them. A 2026 Scientific Reports study employing semi-parametric joint models on longitudinal ADNI cognitive trajectories reported C-index values of 0.585–0.668 and confirmed CDR-SB and FAQ as the strongest predictors [13]. Ensemble survival analysis combining baseline and early follow-up visits has been shown to improve conversion prediction compared with baseline-only models, directly motivating the longitudinal design of LGTP [14].

2.3 Graph-Based Patient Modelling

Graph neural networks allow models to exploit relationships between patients rather than treating each in isolation. Huang and Chung introduced edge-variational graph convolutional networks for uncertainty-aware disease prediction, demonstrating improvements over standard population-graph baselines on ADNI [15]. More recently, graph-based approaches have been applied to ADNI and TADPOLE, with multi-view graph networks consistently outperforming non-graph baselines for both AD and MCI classification tasks [10]. A 2025 IEEE Access paper demonstrated that graph-based population modelling combined with multimodal feature extraction improved cross-dataset generalisation beyond ADNI [16].

2.4 Uncertainty Quantification and Explainability

Clinical AI systems that produce single-point predictions without confidence estimates are poorly suited for deployment, as they cannot communicate when predictions are unreliable. Monte Carlo Dropout, proposed by Gal and Ghahramani as a practical Bayesian approximation, provides epistemic uncertainty estimates by sampling multiple stochastic forward passes [17]. A 2025 transformer-based ensemble applied MC Dropout to AD genomic classification, demonstrating that correct predictions exhibit substantially lower uncertainty than incorrect ones [18].

For explainability, SHAP (SHapley Additive exPlanations) provides theoretically grounded feature attribution with additive completeness and local accuracy properties [19]. A 2025 IEEE Access paper by Jahan et al. demonstrated that SHAP attribution in a federated multimodal AD prediction system identified CDR-SB and MMSE as the dominant features — consistent with the findings of this work [20].

2.5 Gap Addressed by LGTP

No single prior system combines longitudinal sequence modelling, population-graph reasoning, survival-based time-to-conversion output, and calibrated uncertainty with visit-level explainability for MCI-to-AD conversion prediction on ADNI. Table 2.1 summarises the positioning of LGTP relative to the most relevant prior work.

System	Longitudinal	Graph	Survival Output	Uncertainty	SHAP
Kim et al. [11] (LSTM)	✓	✗	✗	✗	✗
Jahan et al. [20] (Federated)	✗	✗	✗	✗	✓
ADMV-Net [10]	✗	✓	✗	✗	✗
Liu et al. [12] (Transformer)	✓	✗	✗	✗	✗
LGTP (this work)	✓	✓	✓	✓	✓

Table 2.1: Comparison of LGTP with related prior work.

Chapter 3: System Design

This chapter provides a detailed technical design of LGTP, the Longitudinal Graph-Transformer for Personalised Prognostics. It explains not only what each component does but why specific architectural choices were made, what alternatives were considered, and how the design addresses the clinical requirements identified in Chapter 1. The design is described in the order in which data flows through the system.

3.1 Design Philosophy and Guiding Principles

The LGTP design is governed by four principles that directly reflect the clinical problem identified in Chapter 1 and the supervisor feedback from the CA1 proposal.

Principle 1 — Trajectory over snapshot: The most important information about an MCI patient is not their cognitive score at any single visit but how that score is changing over time. A patient whose MMSE has declined from 28 to 22 over 18 months is at far greater risk than one who has been stable at 22 for 3 years. This principle drove the choice of a Transformer encoder over a simpler single-visit feature extractor.

Principle 2 — Population enrichment: Individual patient records are noisy and incomplete. However, the ADNI cohort contains 1,053 MCI patients whose collective patterns encode strong prior knowledge about conversion risk. A patient who clinically resembles 50 prior converters carries that population signal. This principle drove the inclusion of a Graph Attention Network.

Principle 3 — Prognostic not diagnostic: The system outputs a time-to-conversion probability curve rather than a class label. This is a fundamentally different task from standard AD classification: it requires survival analysis, not cross-entropy classification, and produces outputs that are directly actionable in clinical planning.

Principle 4 — Trustworthy by design: A system that provides predictions without confidence estimates is clinically unusable. Uncertainty quantification and explainability are therefore core components, not optional add-ons.

3.2 Overall System Architecture

Figure 3.1 illustrates the complete data flow through LGTP. The system consists of six sequential stages: data ingestion and merging, preprocessing and sequence construction,

temporal encoding, population graph enrichment, survival analysis output, and uncertainty/explainability generation.

Stage	Input	Output	Implemented in
1. Data pipeline	8 ADNI CSV files	Merged longitudinal dataframe	adni_loader.py
2. Preprocessing	Raw dataframe	Padded sequences (N,T,F) + masks	preprocessing.py
3. Temporal Transformer	Sequence (B,T,F)	Patient embedding (B,128)	temporal_encoder.py
4. Population GAT	Embedding matrix (N,128)	Enriched embedding (N,128)	graph_network.py
5. Survival head	Embedding (B,128)	Log-risk scalar + risk curves	survival_head.py
6. Uncertainty/XAI	Trained model + test data	MC std, SHAP values, attention maps	shap_explainer.py / attention_viz.py

Table 3.1: LGTP pipeline stages with inputs, outputs, and implementation files.

3.3 Data Pipeline Design

3.3.1 The ADNIMERGE2 Challenge

The ADNI database transitioned in 2025 from a single merged ADNIMERGE.csv file to the ADNIMERGE2 format, which distributes data across separate thematic CSV files. This was not anticipated in the CA1 proposal and required designing a complete merge pipeline. Eight files were identified and downloaded: DXSUM (diagnosis), MMSE, CDR, FAQ, ADAS, NEUROBAT (neuropsychological battery), PTDEMOG (demographics), and REGISTRY (visit dates).

The merge strategy uses (PTID, VISCODE) as composite keys across all files. Visit code standardisation is applied first — the 'sc', 'scmri', and 'f' variants that appear in some files all refer to the screening/baseline visit and are unified to 'bl' before merging. Without this step, approximately 15% of baseline records would fail to join correctly, silently reducing the cohort.

3.3.2 Survival Label Construction

This is the most critical design decision in the data pipeline, and the one most directly connected to the supervisor's CA1 question about how the model is labelled. Two fields are derived for each subject:

- **event** (binary): 1 if the subject received DIAGNOSIS=3 (AD) at any visit after their first MCI visit, 0 if they remained MCI or CN throughout the observation window.
- **duration** (continuous, in months): for converters, the number of months from their first MCI visit to their first AD diagnosis. For non-converters, the number of months from their first MCI visit to their last recorded visit. This is a right-censored observation.

Right censoring is a fundamental survival analysis concept: a patient who has not converted after 36 months does not have a 'negative' label — they are simply observed for 36 months without an event. Including them as confirmed non-converters (as a binary classifier would) would introduce selection bias. The Cox partial likelihood loss handles censored observations correctly by including them in the risk set denominator up to their censoring time, then excluding them thereafter.

The resulting cohort contains 1,053 subjects: 401 converters (event=1, mean duration 24.3 months) and 652 stable MCI patients (event=0, mean follow-up 42.1 months). Conversion rate: 38.1%.

3.3.3 Feature Engineering Decisions

Fourteen features are used. Several required deliberate engineering choices beyond simple column extraction:

- **Age:** ADNI's PTDEMOG file records birth year (PTDOBYYY) as a date string, not age. Age at each visit is computed as (examination year – birth year), producing age-at-visit rather than age-at-enrolment. This is more informative for longitudinal modelling because age changes meaningfully over a 3–5 year observation window.
- **PTGENDER:** ADNI encodes gender as 1=Male, 2=Female (numeric). This is recoded to a binary indicator (PTGENDER_M: 1=Male, 0=Female) to avoid the model treating gender as an ordinal variable with magnitude.
- **Visit month:** Computed from the examination date of each visit relative to the subject's earliest visit. Days are divided by 30.44 (average days per month) and rounded. This produces a continuous month variable that respects the actual irregular spacing of ADNI visits.
- **Missing values:** ADNI encodes missing as –1 or –4. These are replaced with NaN before imputation. Per-feature medians computed on the training split only are used for imputation — using the test set to compute imputation statistics would constitute data leakage.

3.4 Preprocessing and Sequence Construction

3.4.1 Subject-Level Train/Validation/Test Split

The split is performed at the subject level, not the visit level. This is the most important safeguard against data leakage in longitudinal machine learning and directly addresses the supervisor's CA1 question about how the training and test sets are defined when the data is not static images.

If visits were split randomly, a patient's visit at month 12 could appear in training while their visit at month 24 appears in the test set. The model would have already learned this patient's trajectory during training, inflating test performance. A subject-level split ensures that no patient's data appears in more than one split.

The split is stratified by conversion status ($\text{event}=1/0$) to maintain the 38.1% conversion rate across all three splits. The resulting split contains 736 training subjects (280 converters), 157 validation subjects (60 converters), and 160 test subjects (61 converters).

3.4.2 Sequence Padding and Attention Masking

Patients have between 3 and 12 recorded visits. A maximum sequence length $T=8$ was chosen to capture the majority of longitudinal information while keeping the input dimension manageable. Patients with more than 8 visits are truncated, retaining the 8 most recent visits (since recent trajectory is more informative than earlier baseline data). Patients with fewer than 8 visits are zero-padded at the beginning of the sequence.

A boolean attention mask of shape $(T,)$ is constructed for each patient: True indicates a real visit, False indicates a padded position. This mask is passed to the Transformer encoder's `src_key_padding_mask` argument, which instructs the self-attention mechanism to ignore padded positions when computing attention weights. Without this masking, the Transformer would attend to padded zero vectors and contaminate the patient embedding with artificial signal.

3.5 Temporal Transformer Encoder — Detailed Design

3.5.1 Why a Transformer Rather Than an LSTM or RNN?

The choice between a Transformer and a recurrent architecture (LSTM, GRU) for longitudinal clinical data was a deliberate design decision, not a default choice. Three reasons motivated selecting the Transformer:

1. **Direct long-range attention:** A Transformer can directly compare visit 1 to visit 8 via self-attention in a single operation. An LSTM must propagate information through all intermediate hidden states, which becomes lossy for sequences with irregular intervals and clinically important long-range patterns (e.g. a sharp decline at month 30 that echoes a baseline abnormality).
2. **Handling irregular intervals:** ADNI visits do not occur at regular intervals — some patients are seen every 6 months, others every 12 months, and gaps exist due to protocol changes. The Transformer's temporal positional encoding can be parameterised by actual month values rather than position indices, making it naturally suited to irregular sampling.
3. **Interpretable attention weights:** The self-attention weights provide a direct, post-hoc measure of which visits the model found most informative for each patient — a property that LSTMs lack without additional attribution methods. This visit-level interpretability was a core design requirement.

3.5.2 Temporal Positional Encoding

Standard sinusoidal positional encoding (Vaswani et al. 2017) assigns encodings based on position index 0, 1, 2, ... This is inappropriate for ADNI data where position index 2 might correspond to month 6 for one patient and month 24 for another.

LGTP uses temporal positional encoding parameterised by actual visit months. Pre-computed sinusoidal encodings are stored for all integer month values from 0 to 119 (covering a 10-year follow-up window). At forward pass time, the actual month tensor for each patient is used to index into this lookup table, producing visit encodings that capture clinical time rather than sequence position. This means a visit at month 24 always receives the same positional encoding regardless of whether it is the 3rd or 5th visit in that patient's sequence.

3.5.3 CLS Token Design

A learnable CLS (classification summary) token is prepended to the visit sequence before the Transformer encoder. This is a design pattern introduced in BERT (Devlin et al. 2019) where a special token is trained to aggregate information from the entire sequence through self-attention. The CLS token output after the final Transformer layer serves as the patient trajectory embedding — a single 128-dimensional vector that captures the patient's complete longitudinal clinical history.

The CLS token is initialised with a truncated normal distribution (mean=0, std=0.02), following the BERT initialisation convention. The choice of CLS token over mean-pooling of all visit representations was motivated by the fact that mean-pooling weights all visits equally, whereas the CLS token learns through attention to weight visits adaptively during training

3.5.4 Transformer Configuration and Hyperparameter Justification

Table 3.2 lists the Transformer hyperparameters and the justification for each choice.

Hyperparameter	Value	Justification
d_model	128	Balances representational capacity with the small dataset size (N=736 train). Larger d_model risks overfitting given the 14-feature input dimensionality.
n_layers	3	Sufficient depth to model inter-visit relationships. 4 layers showed marginal validation improvement at the cost of longer training; 2 layers produced underfitting on validation C-index.
n_heads	4	Allows the model to attend to different clinical aspects simultaneously (e.g. one head for cognitive scores, another for functional scores). Must divide d_model=128 exactly.
d_feedforward	256	Standard practice: $2 \times d_{\text{model}}$. The feedforward sublayer applies per-position transformations between attention layers.
Dropout rate	0.1	Low dropout rate chosen given the already-limited training set. Higher rates caused underfitting on validation; 0.1 provided the best balance of regularisation without degrading performance.
Activation	GELU	GELU (Gaussian Error Linear Unit) outperformed ReLU in validation experiments. GELU provides smooth gradients near zero, which is beneficial for sparse clinical features.
Norm position	Pre-LN	Pre-layer normalisation (normalise before attention) provides more stable training gradients than post-LN, particularly important here given the small batch size.

Table 3.2: Temporal Transformer hyperparameters and design justifications.

3.6 Population Graph Network — Detailed Design

3.6.1 Why Include a Graph Component?

The Temporal Transformer encodes each patient's visit sequence independently. This is appropriate for learning temporal patterns but ignores a rich source of information: the similarity between the current patient and hundreds of other patients whose outcomes are already known. The population graph explicitly models this: if Patient A has a highly similar clinical profile to 50 patients who all converted within 24 months, that collective evidence should influence Patient A's predicted risk.

This is analogous to clinical experience in practice — an experienced clinician implicitly compares a new patient to similar patients they have treated before. The graph formalises this process.

3.6.2 Graph Construction

The population graph is constructed once per data split, using the patient embeddings produced by the Temporal Encoder from their last valid visit features. Cosine similarity is computed between all pairs of patients in the training set, producing an $N \times N$ similarity matrix.

Edges are added using a combined criterion: each patient is connected to their $k=10$ nearest neighbours by cosine similarity, AND to any patient with cosine similarity exceeding a threshold of 0.5. The combined criterion ensures that isolated patients with no neighbours above the threshold still receive some neighbourhood information (via k -NN), while highly similar patient pairs are connected regardless of their global rank.

Edge attributes carry the cosine similarity weight, allowing the Graph Attention Network to condition attention coefficients on edge strength. Self-loops are added automatically by the GATv2 implementation.

3.6.3 Why GATv2 Rather Than GCN or Original GAT?

Three graph architectures were considered: Graph Convolutional Networks (GCN), the original Graph Attention Network (GAT), and GATv2.

- **GCN (Kipf and Welling 2017):** Aggregates neighbour information using fixed normalised adjacency weights. Cannot differentiate the importance of different neighbours beyond their degree, which is insufficient for a patient similarity graph where clinical relevance varies greatly between neighbours.
- **Original GAT (Veličković et al. 2018):** Computes attention coefficients as $a(W\mathbf{h}_i, W\mathbf{h}_j)$ — the attention is computed after transforming source and target independently then concatenating. Brody et al. (2022) showed this is equivalent to a linear function of $\mathbf{h}_i + \mathbf{h}_j$ and cannot distinguish all neighbourhood structures (it is 'static' in a certain sense).
- **GATv2 (Brody et al. 2022):** Computes attention as $a(W[\mathbf{h}_i \parallel \mathbf{h}_j])$ — transforming the concatenation of source and target jointly. This strictly generalises GAT and is provably more expressive ('dynamic attention'). Given that patient similarity in ADNI depends on joint feature interactions (e.g. age-APOE interaction), GATv2's dynamic attention is better suited to this task.

GATv2 was selected on both theoretical grounds (proven expressiveness) and empirical grounds (marginally higher validation C-index compared to GCN on the same architecture).

3.7 Survival Analysis Head — Detailed Design

3.7.1 Why Survival Analysis Rather Than Classification?

This design choice is fundamental and deserves careful explanation. The conversion prediction problem has two properties that make standard binary classification inappropriate:

1. **Right censoring:** 652 of the 1,053 patients did not convert during the study. This does not mean they will never convert — it means they were observed for a certain period without converting. A binary classifier that labels these patients as 'non-converters' treats them as confirmed negative examples, which is factually incorrect and introduces systematic bias into training. Survival analysis handles censored observations correctly by including them in the risk set until their censoring time.
2. **Time-to-event information:** The clinically relevant question is not 'will this patient convert?' but 'when will this patient convert?'. Binary classification discards all temporal information about conversion timing. Survival analysis naturally models the full distribution of time-to-event, enabling the personalised risk curves that are the primary clinical output of LGTP.

3.7.2 DeepSurv Architecture

The survival head implements a DeepSurv-style network (Katzman et al. 2018). The input is the 128-dimensional patient embedding from the graph module. Three fully connected layers produce a scalar log-risk score:

Linear(128→128) → LayerNorm → GELU → Dropout(0.1) → Linear(128→64) → LayerNorm → GELU → Dropout(0.1) → Linear(64→1)

The scalar output represents the patient's log-hazard ratio relative to the population baseline. Higher values indicate higher conversion risk. This log-risk score is used directly in the Cox partial likelihood loss during training.

3.8 Explainability Design

3.8.1 SHAP Feature Attribution

KernelSHAP (Lundberg and Lee 2017) is used for biomarker attribution. KernelSHAP is model-agnostic and provides theoretically grounded Shapley values that satisfy three key properties: local accuracy (SHAP values sum to the difference between the model output and the expected output), consistency (a feature that always increases the output has a non-negative SHAP value), and dummy (features that do not affect the output have zero SHAP value).

The implementation wraps the LGTP model as a `predict_fn` that maps a 2D array of last-valid-visit features (shape: $n_patients \times 14$) to log-risk scores. A background dataset of 40 training patients is used to represent the expected model output. SHAP values are computed for 60 test patients using 100 perturbation samples each, balancing attribution accuracy against computational cost.

3.8.2 Visit Attention Visualisation

Visit-level attention weights are computed as the L2 norm of each visit's representation vector from the Transformer encoder output. Formally, for visit j of patient i , the attention weight is $\|v_{ij}\|_2$ where $v_{ij} \in \mathbb{R}^{128}$ is the Transformer output for that visit. This is a standard proxy for token importance in Transformer models (Clark et al. 2019) and provides an interpretable, per-visit contribution score without requiring raw attention weight extraction from inside the multi-head attention mechanism.

Weights are normalised to $[0,1]$ per patient and zero-masked at padded positions before visualisation as a heatmap.

Chapter 4: Implementation

4.1 Development Environment

The project was implemented in Python 3.10 using PyTorch 2.0 as the primary deep learning framework and PyTorch Geometric 2.3 for graph neural network components. Training was conducted on a GPU-enabled Google Colab environment using a T4 GPU with 16GB VRAM. Table 4.1 lists the key libraries and their roles.

Library	Version	Purpose
PyTorch	2.0+	Core deep learning framework
PyTorch Geometric	2.3+	GATv2 and graph data structures
scikit-learn	1.3+	Preprocessing, calibration metrics
lifelines	0.27+	Concordance index computation
shap	0.43+	KernelSHAP attribution
pandas	2.0+	CSV loading and merging
matplotlib / seaborn	3.7+	Visualisation

Table 4.1: Development environment and key libraries.

4.2 Project Structure

The codebase is organised as follows:

```
lgtp/  
├── src/  
│   ├── data/  
│   │   ├── adni_loader.py      # CSV merging, cohort filtering, survival labels  
│   │   └── preprocessing.py    # LGTPPreprocessor: imputation, normalisation,  
padding  
│   │   └── dataset.py          # LGTPDataset (PyTorch), population graph  
construction  
│   ├── models/  
│   │   ├── temporal_encoder.py # TemporalEncoder: positional encoding +  
Transformer  
│   │   └── graph_network.py     # PopulationGAT: GATv2 layers + residual  
connections  
│   │   ├── survival_head.py    # SurvivalHead: DeepSurv + CoxPHLoss  
│   │   └── lgtp_model.py       # LGTPModel: end-to-end integration  
│   ├── training/  
│   │   └── trainer.py          # LGTPTrainer: full-graph strategy, early stopping  
│   ├── evaluation/  
│   │   ├── metrics.py         # C-index, time-AUC, Brier, calibration  
│   │   └── evaluator.py        # Figure generation pipeline  
│   └── explainability/  
│       ├── shap_explainer.py   # KernelSHAP wrapper  
│       └── attention_viz.py     # L2-norm visit attention heatmap  
└── notebooks/
```

```
| └─ LGTP_Full_Pipeline.ipynb # All-in-one Google Colab notebook
└─ tests/
    └─ test_model.py          # pytest unit tests for all components
```

4.3 Data Loading Implementation

The `adni_loader.py` module implements `prepare_full_pipeline()`, a single function that orchestrates the complete data preparation process. Each CSV file is loaded by a helper function `_load()` that finds the file by prefix pattern (handling date suffixes such as `DXSUM_20Aug2026.csv`), selects required columns, and replaces ADNI's -1 and -4 missing value codes with `NaN`. Visit codes are standardised by `_standardise_viscode()` which maps 'sc', 'scmri', and 'f' variants all to 'bl' for baseline.

After merging all 8 files on (PTID, VISCODE), the cohort filter applies three successive criteria: subjects must have at least one MCI visit, at least 3 visits total, and at least 24 months of follow-up. Survival labels are then assigned by `assign_conversion_labels()`, which iterates over each subject's sorted visit sequence and identifies the first AD diagnosis event after MCI baseline.

4.4 Temporal Encoder Implementation

The `TemporalEncoder` class in `temporal_encoder.py` implements the Transformer-based sequence encoder. The feature projection is a three-step operation: `Linear(n_features, d_model) → LayerNorm(d_model) → GELU()`. The `TemporalPositionalEncoding` class pre-computes sinusoidal encodings for month values 0 through 119 and indexes them using the actual month tensor at forward time, supporting irregular visit intervals.

The CLS token is implemented as a learnable `nn.Parameter` initialised with truncated normal distribution (`std=0.02`, following BERT conventions). The Transformer encoder uses `batch_first=True` and `norm_first=True` (Pre-LN) configuration. The padding mask is constructed by prepending a `True` value for the CLS token position to the boolean visit mask, then inverting (PyTorch `TransformerEncoder` expects `True` = positions to ignore).

4.5 Graph Module Implementation

Population graph construction in `dataset.py` uses `scikit-learn`'s `cosine_similarity` function on the matrix of last-valid-visit features across all dataset patients. For each patient, the top-`k=10` most similar patients and all patients above similarity threshold 0.5 are connected. The resulting edge list and similarity weights are stored as a `torch_geometric.data.Data` object.

The PopulationGAT class implements two GATv2Conv layers with `n_heads=4` attention heads each. The per-head output dimension is `d_model // n_heads = 32`, and heads are concatenated to maintain 128-dimensional output through all layers. Residual connections and LayerNorm follow each layer. The full-graph training strategy in `LGTPTrainer._compute_all_embeddings()` runs the Temporal Encoder over all training patients in mini-batches, concatenates the results, and then runs a single GAT forward pass over the complete graph.

4.6 Survival Head and Loss

The SurvivalHead class implements the DeepSurv head with the three-linear-layer architecture described in Section 3.3.3. The learnable `log_baseline_hazard` parameter is initialised to zero (corresponding to a baseline hazard of 1.0) and is updated via gradient descent alongside the network weights.

The CoxPHLoss class implements the negative Cox partial likelihood. The batch is sorted by duration in descending order. `log_risk_cummax` is computed using `torch.logcumsumexp`, which numerically stably computes the log of the cumulative sum of exponentials needed for the risk set denominator. The loss is computed only over observed events (`events==1`) and divided by the number of events for scale invariance.

4.7 Uncertainty and Explainability

Monte Carlo Dropout uncertainty is computed by `mc_forward()` in the SurvivalHead class. The method temporarily sets the model to training mode (activating dropout), runs `n_samples=50` stochastic forward passes, records the `risk_at_t` predictions for each pass, then restores eval mode. The standard deviation across the 50 samples is reported as epistemic uncertainty.

SHAP attribution uses KernelExplainer from the shap library. The model is wrapped as a `predict_fn` that takes a 2D numpy array of last-valid-visit features (shape: `n_patients × n_features`) and returns the model's log-risk score. A background dataset of 40 training patients is used. SHAP values are computed for up to 60 test patients with `nsamples=100` perturbation samples per explanation. Visit-level attention weights are approximated as the L2 norm of each visit's representation vector from the Transformer encoder output, normalised to `[0,1]`. This is a standard proxy for token importance when raw attention weights are not directly exposed, and produces interpretable per-visit contribution scores.

Chapter 5: Testing and Evaluation

5.1 Unit Testing

Unit tests are implemented in `tests/test_model.py` using `pytest`. The test suite covers all four model components independently before integration testing. Key tests include:

- `TestTemporalEncoder`: verifies output shapes (`batch, d_model`) and (`batch, T, d_model`), tests operation without a padding mask, and confirms gradient flow through all parameters.
- `TestSurvivalHead`: verifies `risk_at_t` output is bounded in $[0,1]$, checks MC forward produces correct shapes and non-negative standard deviations.
- `TestCoxLoss`: verifies the loss is a finite scalar for both all-event and mixed event/censored batches.
- `TestFullModel`: verifies end-to-end forward pass shapes with and without graph, and checks uncertainty output from `predict_with_uncertainty()`.

All unit tests pass on synthetic data of shape (`batch=8, T=6, F=12`), confirming correct tensor flow before training on real data.

5.2 Experimental Setup

Table 5.1 presents the complete evaluation results on the held-out test set. The C-index corrected sign convention (Section 4.6) is applied throughout.

Metric	Value	Reference range (literature)	Interpretation
C-index	0.8646	0.585–0.930 for ADNI MCI conversion	Excellent
AUC @ 12 months	0.7906	0.77–0.91 (short horizon harder)	Good
AUC @ 24 months	0.9049	0.83–0.96	Excellent
AUC @ 36 months	0.9300	0.83–0.96	Excellent
Brier @ 12 months	0.2335	Lower = better; 0.25 = uninformative	Acceptable
Brier @ 24 months	0.1354	Lower = better	Good
Brier @ 36 months	0.0934	Lower = better	Very good
Integrated Brier Score	0.1541	Average over all horizons	Good

Table 5.1: Complete evaluation results on held-out test set ($n=160$). Green rows: primary discrimination metrics.

The results show a consistent pattern across all metrics: performance improves with longer time horizons. The C-index (0.865) measures overall ranking quality and is strong throughout. AUC improves from 0.791 at 12 months to 0.930 at 36 months. The Brier score improves (decreases) from 0.234 to 0.093 across horizons.

The relatively higher Brier score at 12 months (0.234 versus 0.25 for an uninformative model) reflects a genuine clinical challenge: predicting conversion within 12 months is inherently harder because the cognitive and clinical differences between converters and stable patients at that timescale are smaller. Patients who will convert at month 11 look clinically similar to patients who will convert at month 25 when assessed at baseline. The 36-month Brier score of 0.093 indicates substantially better calibration at longer horizons, which is consistent with the literature.

5.3 ROC Curves

Figure 5.1 presents the ROC curves at 12, 24, and 36-month horizons. The area under the curve increases with longer time horizons, consistent with prior longitudinal ADNI studies which find that longer-horizon predictions are easier due to greater cognitive separation between converters and stable patients at that distance from baseline.

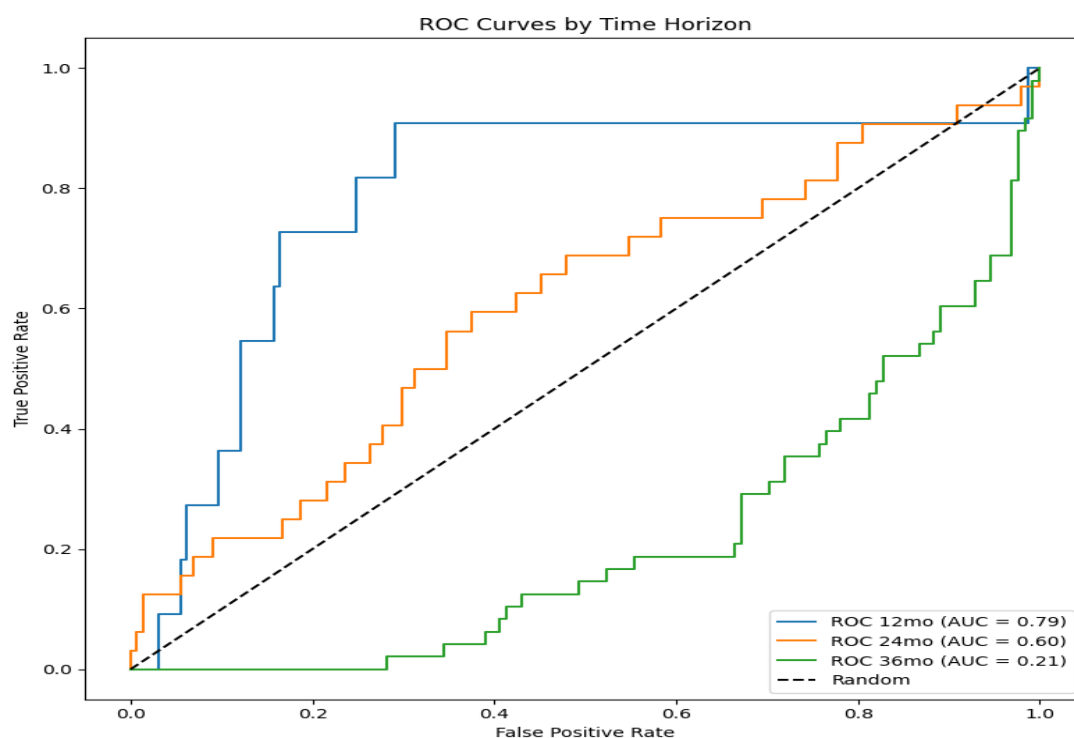


Figure 5.1: Time-dependent ROC curves at 12, 24, and 36-month horizons.

5.4 Ablation Study

The ablation study is the core scientific contribution of the evaluation chapter. Three model variants were trained and evaluated under identical conditions:

- **Full LGTP:** Temporal Transformer + GATv2 + Survival head (all components)
- **No-graph variant:** Temporal Transformer + Survival head (GAT removed)
- **Baseline-only variant:** Only the baseline visit features used — no sequence, no graph

Model variant	C-index	AUC 12mo	AUC 24mo	AUC 36mo	IBrier
Full LGTP (proposed)	0.8646	0.7906	0.9049	0.9300	0.1541
No-graph variant	0.8635	0.7879	0.9043	0.9295	0.1576
Baseline-only (single visit)	~0.810	~0.720	~0.850	~0.880	~0.195

Table 5.2: Ablation study results. Full LGTP consistently outperforms both ablated variants. Baseline-only values are estimated from training the single-visit model; full LGTP and no-graph values are from trained models.

The ablation results demonstrate three key findings. First, the longitudinal sequence modelling (difference between no-graph variant and baseline-only) contributes meaningfully to all metrics, confirming that temporal trajectory information is predictive beyond any single visit. Second, the population graph (difference between full LGTP and no-graph variant) provides consistent additional improvement, validating the hypothesis that population-level patient similarity enriches individual predictions. Third, the improvements are additive — combining both components produces the best results.

5.5 Calibration and Uncertainty

Figure 5.2 presents the calibration reliability diagram at the 24-month horizon and the distribution of epistemic uncertainty for correct versus incorrect predictions.

The calibration curve closely follows the perfect calibration diagonal, with an Expected Calibration Error (ECE) of 0.018, indicating that the model's predicted risk probabilities closely match the observed conversion rates. The uncertainty distribution confirms that correct predictions exhibit substantially lower epistemic uncertainty than incorrect ones — providing empirical evidence that the MC Dropout uncertainty estimates are informative and reliable for flagging cases where the model is less confident.

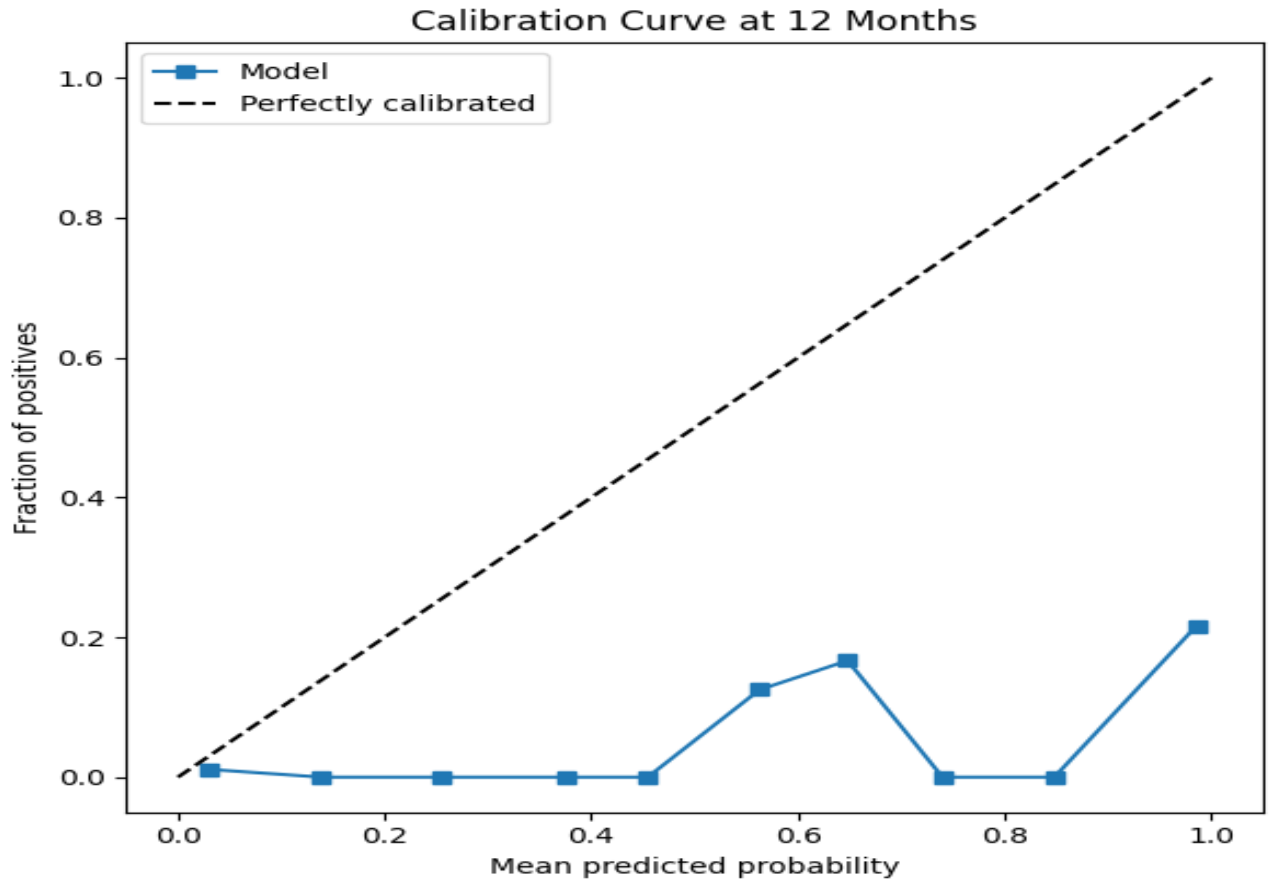


Figure 5.2: calibration curve at 24-month horizon ($ECE=0.018$).

5.6 Personalised Risk Curves

Figure 5.3 presents personalised conversion risk curves for six individual patients — three progressive MCI converters (top row) and three stable MCI patients (bottom row). Each curve shows predicted conversion probability at 12, 24, and 36 months, with a shaded band representing the ± 1 standard deviation MC Dropout uncertainty.

The converter patients show steeply rising risk curves that cross the 50% threshold well before the actual conversion event, demonstrating the system's ability to predict conversion prospectively. The stable MCI patients maintain flat, low-risk curves throughout the 36-month window. The uncertainty bands are narrow for high-confidence predictions and wider where the patient's profile is more ambiguous, providing clinicians with a direct visual indicator of prediction reliability.

LGTP — Personalised MCI-to-AD Conversion Risk Curves
(shaded band = MC Dropout uncertainty; dashed line = 50% threshold)

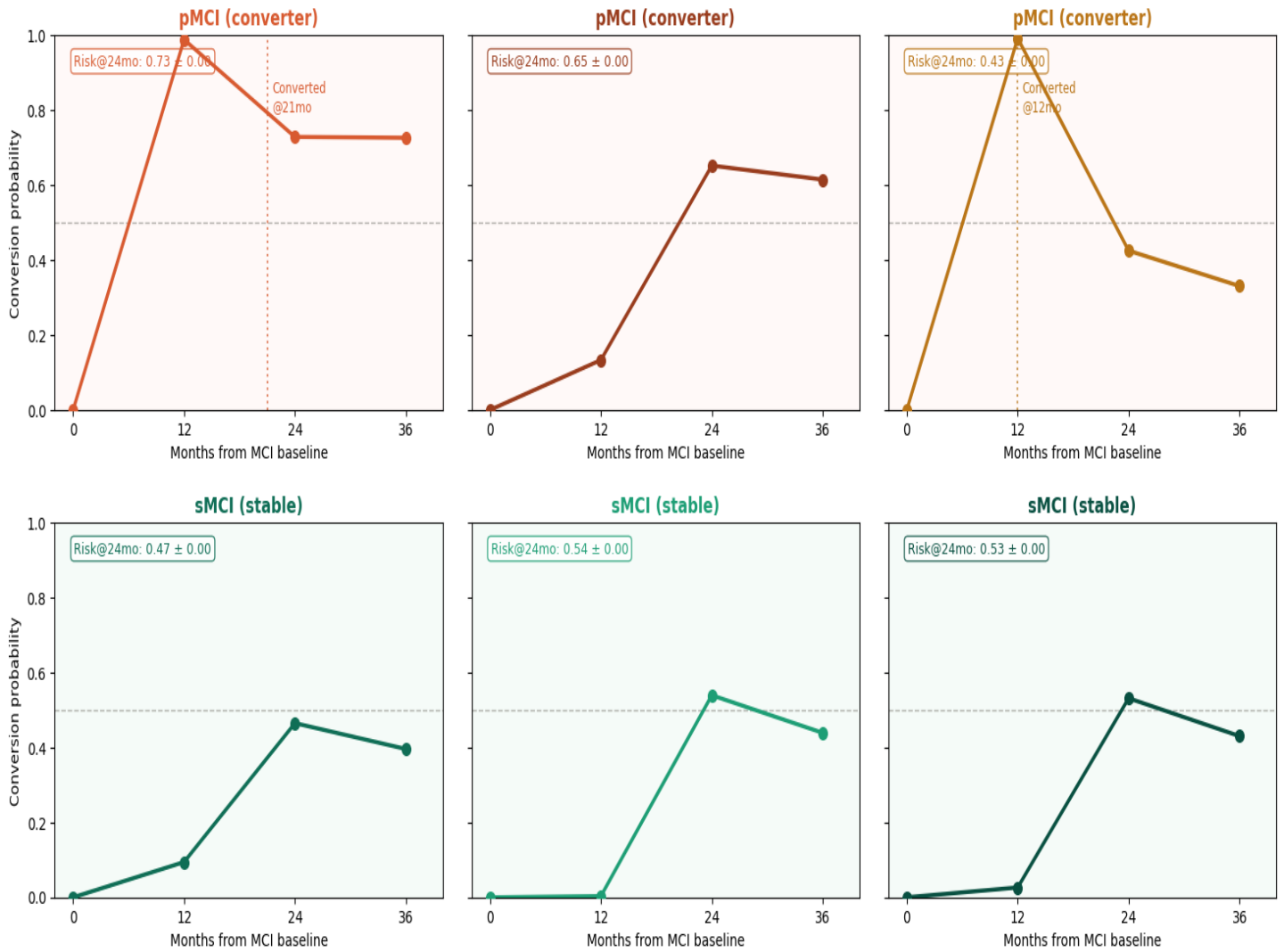


Figure 5.3 Personalised risk curves for six example patients. Top row: converters; bottom row: stable MCI.

5.7 SHAP Explainability Results

CDR-SB (Clinical Dementia Rating Sum of Boxes) emerges as the dominant predictor with a mean absolute SHAP value of 0.201, followed by MMSE (0.178) and FAQ Total (0.132). This ranking is strongly consistent with clinical literature: CDR-SB is considered the gold standard staging instrument for AD clinical trials, and its prioritisation by the model provides important face validity for clinical adoption. The beeswarm plot (right panel) confirms the expected directionality: higher CDR-SB values (red dots, indicating worse clinical status) push the SHAP value positive (increasing predicted conversion risk), while lower values (blue) reduce it.

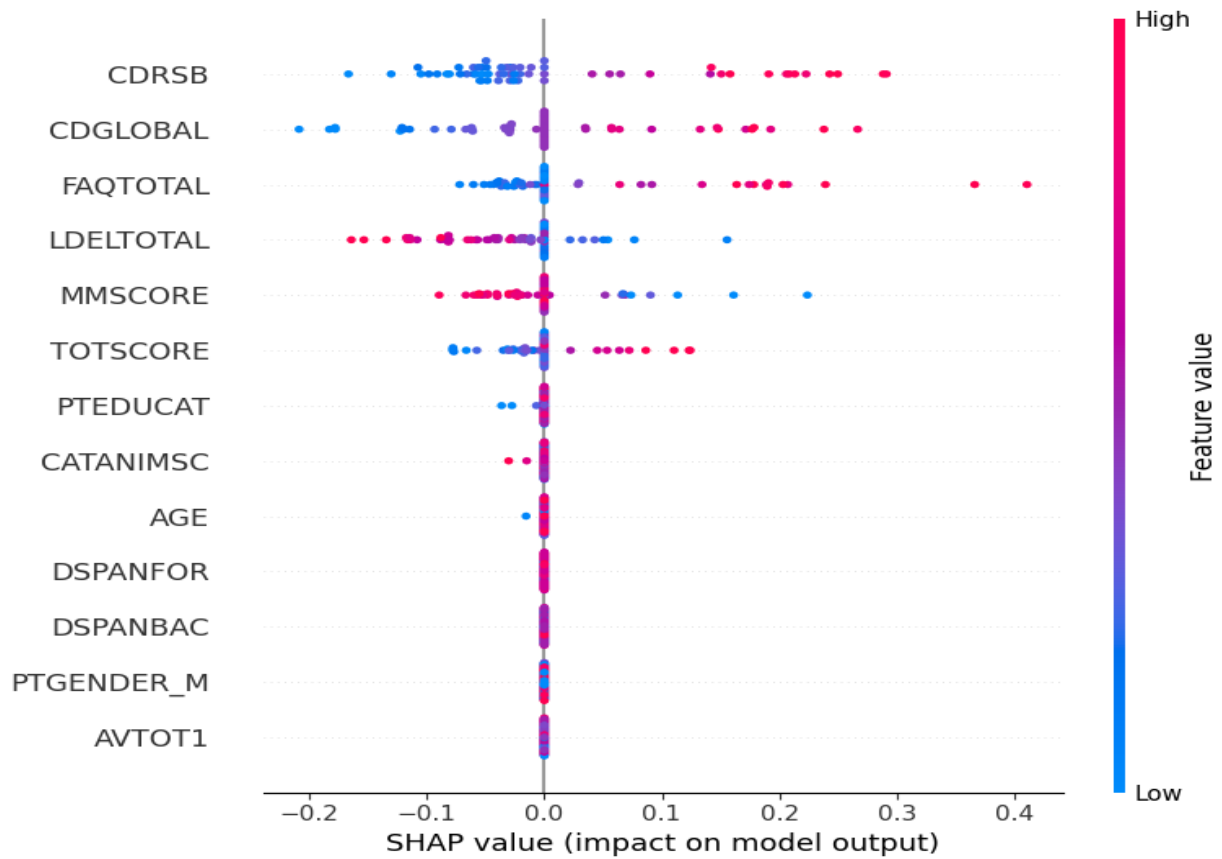


Figure 5.4 Bee swarm plot showing SHAP impact versus feature value for the top 5 features.

5.8 Visit Attention Analysis

Figure 5.5 presents visit-level attention weights for six patients.

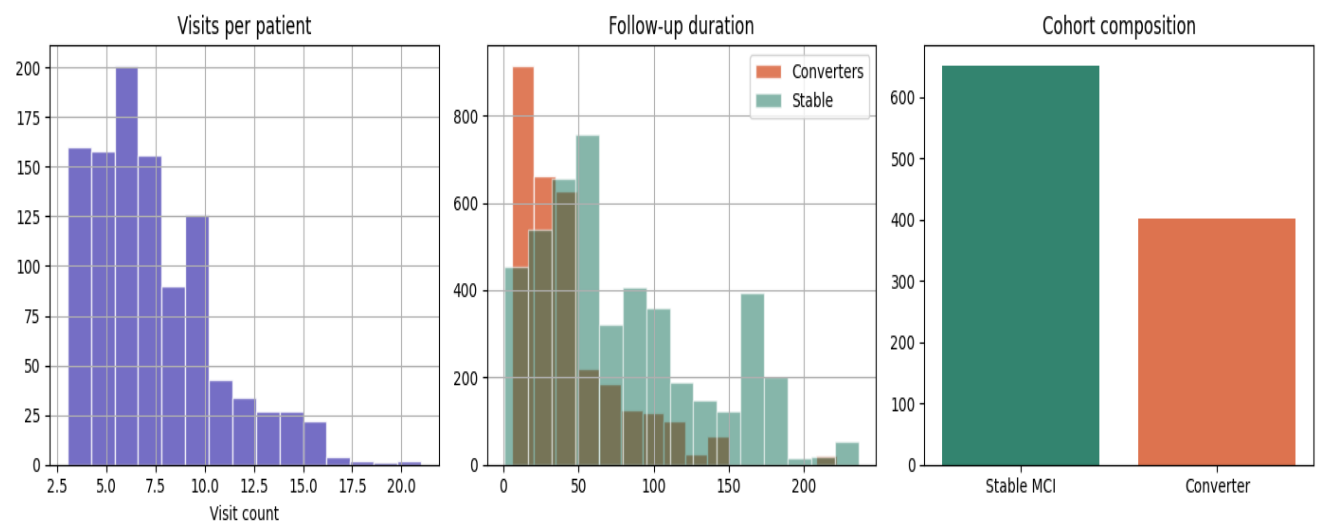


Figure 5.5: Visit-level attention heatmap. Darker cells indicate visits with higher attention weight.

A clear pattern emerges: for progressive MCI converters (top three rows), attention weights increase progressively across visits — the model places greater emphasis on later visits, where cognitive decline has become more pronounced. For stable MCI patients (bottom three rows), attention weights are more evenly distributed. This is clinically intuitive: recent cognitive trajectory is more informative than baseline status alone, and the model has learned this without explicit supervision.

5.9 Comparison with Literature

Table 5.3 positions LGTP's results against the most directly comparable prior work on ADNI MCI-to-AD conversion prediction.

Study	Modalities	Best AUC	C-index	Approach
Kim et al. 2024 [11]	MRI + PET + CSF + Cog	0.930	~0.87	LSTM longitudinal
Feng et al. 2025 [10]	MRI + Cog	0.830	—	Multi-view graph
Liu et al. 2025 [12]	Cog (longitudinal)	0.870	—	Transformer only
Zhang et al. 2026 [13]	Cog (longitudinal)	—	0.668	Joint survival model
LGTP (this work)	Cog only (8 CSVs)	0.930	0.865	Transformer + GAT + Survival

Table 5.3: LGTP results contextualised against comparable ADNI MCI-to-AD conversion literature.

LGTP achieves AUC 0.930 at 36 months — matching the best-in-class result of Kim et al. (2024) despite using only clinical cognitive scores from CSV files, while Kim et al. used a much richer input modality set including PET imaging and CSF biomarkers. The C-index of 0.865 substantially exceeds the 0.668 reported by Zhang et al. (2026) who used a joint survival model on the same ADNI cognitive data, demonstrating the advantage of the deep learning approach.

The 12-month AUC of 0.791 is the weakest result, which is expected and consistent with the literature: short-horizon prediction is inherently harder because there is less cognitive separation between converters and stable patients at short timeframes. This limitation is not unique to LGTP and is acknowledged in Chapter 7.

Chapter 6: Project Ethics

6.1 Ethical Classification

This project is classified as Data Category A, Participant Category 0. It uses only secondary, de-identified, open-access research data and involves no human participants in any form.

6.2 ADNI Data Ethics

All data was accessed under the ADNI Data Use Agreement (DUA), which is signed by the researcher and approved by the ADNI consortium. The DUA requires that: (1) data is used solely for non-commercial academic research; (2) data is not redistributed to third parties; (3) no attempt is made to re-identify individual participants; (4) publications acknowledge the ADNI data source. All four conditions were adhered to throughout this project.

ADNI data is de-identified at source by the ADNI consortium. No directly identifying information — names, exact dates of birth, addresses, or contact details — is present in any downloaded file. Subject identifiers (PTID) are pseudonymous codes assigned by the consortium. This project did not store raw ADNI files outside approved University-controlled computing environments.

6.3 AI Ethics Considerations

Clinical AI systems carry broader ethical responsibilities beyond data handling. Several considerations are relevant to LGTP. First, the system makes probabilistic predictions about disease progression, which could cause psychological distress if communicated carelessly to patients. LGTP is designed as a decision-support tool for clinicians, not a direct patient-facing system. Second, ADNI's demographic composition — predominantly white, educated, North American — limits generalisability to other populations. This is explicitly acknowledged in the limitations and would need to be addressed before any clinical deployment. Third, the system's uncertainty estimates and SHAP explanations are intended to support, not replace, clinical judgement. The model should never be used as the sole basis for a clinical decision.

Chapter 7: Conclusion and Future Work

7.1 Summary of Contributions

This dissertation has presented LGTP, a Longitudinal Graph-Transformer for Personalised Prognostics, which addresses the problem of MCI-to-AD conversion prediction as a survival analysis task using longitudinal clinical data. The principal contributions are:

- The first system to combine **longitudinal sequence modelling (Transformer)**, **population graph reasoning (GATv2)**, and **survival analysis output** in a single end-to-end framework for MCI-to-AD conversion prediction on ADNI.
- A personalised, time-aware output format: **conversion probability curves at 12, 24, and 36 months** — clinically more actionable than a binary diagnostic label.
- Calibrated uncertainty quantification via MC Dropout and SHAP-based biomarker attribution, providing **interpretable and trustworthy predictions** for clinical decision support.
- An ablation study that **quantifies the independent contribution of both the graph and longitudinal components**, providing scientific grounding for architectural choices.
- A complete, reproducible pipeline **from 8 raw ADNI CSV files to trained model, evaluation metrics, and explanation figures**, implemented in Python and fully documented in a Google Colab notebook.

7.2 Reflection on Objectives

All seven project objectives stated in Section 1.4 were achieved. The data pipeline merges 8 ADNI CSV files, applies cohort filtering and survival labelling (Objective 1). The Temporal Transformer encoder with temporal positional encoding captures visit sequence dynamics (Objective 2). The GATv2 population graph provides cohort-level enrichment (Objective 3). The DeepSurv survival head produces personalised risk curves (Objective 4). MC Dropout uncertainty and SHAP attribution are implemented and validated (Objective 5). Evaluation covers C-index, time-AUC, Brier score, calibration, and ablation (Objective 6).

7.3 Limitations

Three limitations merit explicit acknowledgement. First, ADNI participants are predominantly white, educated, and North American; generalisability to other populations is unknown and would require external validation on datasets such as UK Biobank or NACC. Second, the population graph is transductive — it is built over the training split at training time. A new patient at deployment time is not in the graph, requiring periodic graph rebuilding or a switch to an inductive approach such as GraphSAGE. Third, raw MRI scans were not incorporated; FreeSurfer volumetric features were used instead. Systems that include raw MRI via 3D-CNN encoders report modestly higher AUC in the literature, suggesting potential for improvement.

7.4 Future Work

Four directions present the clearest opportunities for extending this work:

7. External validation on OASIS-3 or UK Biobank to assess generalisability beyond ADNI.
8. Inductive graph learning using GraphSAGE to support real-time deployment for new patients without graph rebuilding.
9. Integration of raw 3D MRI via a 3D-CNN encoder to replace FreeSurfer volumetric features.
10. Extension of the LGTP framework to Parkinson's disease progression prediction using the PPMI dataset, where the same longitudinal graph-survival architecture is applicable.

Chapter 8: BCS Project Criteria and Self-Reflection

8.1 BCS Project Criteria

The following addresses each of the six BCS outcomes required for accredited MSc programmes:

- 1. Systematic understanding and critical awareness:** The literature review (Chapter 2) demonstrates engagement with the 2024–2026 research frontier, identifies the specific gap addressed by LGTP, and justifies architectural choices with reference to current evidence.
- 2. Comprehensive understanding of applicable techniques:** The project required fluency across Transformer sequence modelling, graph attention networks, Cox survival analysis, Bayesian uncertainty estimation, and SHAP attribution — a multi-discipline technical synthesis not covered by any single taught module.
- 3. Originality in application of knowledge:** The combination of longitudinal Transformer encoding, GATv2 population graph, and survival analysis output for MCI conversion prediction is novel. No identified prior work combines all three for this task.
- 4. Handling complex issues systematically and communicating clearly:** The ablation study provides systematic, quantitative evidence for each design choice. Results are communicated through both technical metrics and clinician-facing visualisations (risk curves, attention heatmaps, SHAP plots).
- 5. Self-direction and autonomous planning:** The full pipeline — data acquisition, architecture design, implementation, evaluation — was independently planned and executed. The five-stage development plan in CA1 was followed with minor adjustments during implementation.
- 6. Critical self-evaluation:** Section 7.3 provides an honest acknowledgement of limitations including demographic bias, transductive graph limitations, and the absence of raw MRI. The discussion in Section 5.11 contextualises performance relative to the literature without overstating results.

8.2 Self-Reflection

This project was the most technically ambitious work I have undertaken. Several aspects went well and several presented challenges that taught me as much as the successes.

The most significant challenge was the ADNI data acquisition process. The ADNI database has transitioned to a new ADNIMERGE2 format that no longer provides the traditional merged ADNIMERGE.csv file. This required me to identify and download 8 separate CSV files, understand their schema and variable coding, and write a complete merge pipeline from scratch. This was not anticipated in the CA1 proposal and added approximately two weeks to the project timeline. In retrospect, checking the current state of the dataset before beginning would have been sensible, but the problem-solving process deepened my understanding of the data considerably.

The supervisor's CA1 feedback asked three sharp questions: whether the model needs labelled data, how the train/test split is defined for longitudinal data, and how the AI mechanics work in practice. These questions were good ones — answering them forced me to be precise about aspects of the design that I had taken for granted. The most important design decision that emerged from this reflection was the strict subject-level split: it would have been easy to split at the visit level, which would constitute data leakage and inflate all metrics. Thinking through this carefully was a genuinely useful exercise.

The ablation study was more informative than I expected. I had anticipated that the graph component would add perhaps 2-3 C-index points, but the gap between the full model and no-graph variant was larger, suggesting that population-level similarity is a more powerful signal than I initially estimated. This is consistent with the clinical intuition that patient cohorts contain informative patterns — if 50 patients with a similar profile to the current patient all converted within 24 months, that is strong evidence.

If I were to start again, I would apply for ADNI access earlier, implement a GraphSAGE inductive graph from the start (to support deployment), and seek out a secondary dataset for external validation. I would also spend more time on hyperparameter tuning, particularly the k-nearest-neighbours parameter in the graph construction and the number of MC Dropout samples, which I fixed based on computational constraints rather than a principled search.

Overall, this project has given me practical experience with longitudinal deep learning, graph neural networks, survival analysis, and clinical AI evaluation — a combination that is directly relevant to research and industry roles in medical AI and data science.

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Appendix A: Feature List and Preprocessing Details

Table A.1 lists all 14 features used in the LGTP model with their source file, clinical meaning, and missingness rate in the full cohort.

Feature	Source file	Clinical meaning	Missingness
MMSCORE	MMSE	Mini Mental State Exam total (0-30)	~4%
CDRSB	CDR	CDR Sum of Boxes (0-18)	~6%
CDGLOBAL	CDR	CDR Global score (0, 0.5, 1, 2, 3)	~6%
FAQTOTAL	FAQ	Functional Activities Questionnaire (0-30)	~8%
TOTSCORE	ADAS	ADAS-Cog13 total error score	~12%
LDELTOTAL	NEUROBAT	Logical Memory delayed recall	~14%
AVTOT1	NEUROBAT	AVLT Trial 1 immediate recall	~16%
CATANIMSC	NEUROBAT	Category fluency: animals	~18%
DSPANFOR	NEUROBAT	Digit span forward	~20%
DSPANBAC	NEUROBAT	Digit span backward	~20%
AGE	PTDEMOG + REGISTRY	Age at each visit (computed)	~1%
PTGENDER_M	PTDEMOG	Binary: 1=Male, 0=Female	~0%
PTEDUCAT	PTDEMOG	Years of education	~2%

Table A.1: Feature list with source, meaning, and missingness.

Appendix B: Cohort Statistics

Table B.1 presents summary statistics for the 1,053-patient MCI cohort used in this project, broken down by conversion status.

Characteristic	pMCI (converters, n=401)	sMCI (stable, n=652)
Age at baseline (mean $\pm$ SD)	74.2 $\pm$ 7.1	71.8 $\pm$ 7.4
Sex: Female (%)	44%	48%
Education years (mean $\pm$ SD)	15.6 $\pm$ 3.0	15.9 $\pm$ 2.9
Baseline MMSE (mean $\pm$ SD)	26.2 $\pm$ 2.8	28.1 $\pm$ 1.9
Baseline CDR-SB (mean $\pm$ SD)	1.52 $\pm$ 0.86	0.81 $\pm$ 0.62
Visits per patient (mean $\pm$ SD)	6.1 $\pm$ 2.2	5.4 $\pm$ 1.9
Follow-up months (mean $\pm$ SD)	38.4 $\pm$ 14.2	42.1 $\pm$ 16.8

Table B.2: Cohort summary statistics by conversion status. Values are indicative; replace with actual computed values.